

Discrete Event Simulation Method to Model Passenger Processing at an International Airport

Kulunu Munasingha

*Department of Transport and Logistics Management
University of Moratuwa
Katubedda, Sri Lanka
kulunu1994@gmail.com*

Varuna Adikariwattage

*Department of Transport and Logistics Management
University of Moratuwa
Katubedda, Sri Lanka
varunaa@uom.lk*

Abstract—The research paper describes a methodology to model departure passenger processing in a terminal at an international airport. This methodology is based on discrete event simulation theory which has the capability to model a network of processors under time varying demand pattern observed in airport terminals and reveal any existing bottlenecks. The discrete event simulation model was developed using Simio 11 software which can be used to do “what-if analysis” of the system. Further, the simulation model was developed based on a case study at Bandaranaike International Airport, Sri Lanka.

Keywords—simulation, waiting time, service time, queue length

I. INTRODUCTION

The air transportation industry shows rapid growth since the last decade. According to the ICAO statistics, 4.3 billion passengers were carried through the air in 2018. When it comes to the Asia Pacific region, the passenger growth was 7.3 % in 2018 compared with 2017(IATA,2018). Therefore, airlines and airports have to play a major role to cater to this ever increasing passenger demand. Nowadays airports are struggling with capacity problems. Most of the airports are unable to balance their demand and capacity with the limited number of infrastructure facilities and human resources. When balancing demand and capacity, main factors that need to be considered are cost, operational efficiency, and customer satisfaction[1].

Nowadays, airports are engaged in massive development to expand their passenger terminals due to passenger growth, tightening security against terrorist attacks, strict security restrictions, and changes in operational policies. Thus, airports need huge facilities to accommodate these essential operations. Apart from that, effective and efficient usage of resources is a mandatory requirement to provide a better service economically. Airport terminals need to handle mainly three types of passenger flows such as originating transfer and terminating. Departure process is the most demanding compared to other types of passenger flow in terms of airport resource utilization. When it comes to the departure process, baggage screening and check-in processes are the most critical since, waiting times, long queues, and initial delays occur if these processes are not efficient.

It is important to study passenger flow at an airport to identify prevailing bottlenecks and suggest corrective actions. Several studies have been done related to passenger flow modeling in airports. These studies equally make use of analytical and simulation techniques[2]. However, only a

limited number of studies attempt to model the complete path of departure or arrival passengers. Modeling the complete flow path provides insights about the interaction between the sequential processes. Thus, it is an effective way of addressing bottlenecks by properly balancing capacity and demand along the entire flow path. Apart from that, airport operators can utilize passenger flow modeling to obtain a microscopic view on each terminal process and their interactions. Thus, precise interventions from both demand and supply aspects can be made towards the operational efficiency of the airport with minimum investment on infrastructure. The methodology used to model passenger processing is a discrete event simulation(DES) theory. Through that, the departure terminal system of a mid-size airport was modeled. The developed model was useful to analyze different scenarios or interventions. Also, the model was used to observe how those scenarios impact on the overall passenger flow.

II. CASE STUDY AT BANDARANAIKE INTERNATIONAL AIRPORT

Bandaranaike International Airport (IATA: CMB, ICAO: VCBI) is the main international airport in Sri Lanka (hereafter referred to as BIA). BIA phase two expansion project was expected to initiate in 2017 and expected to be completed by 2020. But expansion has not been completed on time. Therefore, at the end of 2017, the airport’s existing terminal handled over 10 million passengers per annum. However, it was designed to handle 6 million passengers per annum. As a result of that passengers are facing a lot of difficulties. Due to exceeding capacity, frequent instances of overcrowding and as a result, customer satisfaction has been affected. Therefore, it is necessary to model existing passenger flow at BIA to identify existing bottlenecks and make recommendations to increase the operational efficiency of the passenger handling process.

III. LITERATURE REVIEW

When it comes to airport landside models, the classification of them is based on two essential characteristics in demand variation such as stochasticity and time variation. Under time variation, models take into account the demand variation for various facilities at the airport landside depending on the time of day. Under stochasticity, random variation of demand or the stochastic variation of demand rate within a limited period of time is considered. So according to those time variations and stochasticity, four models can be defined as follows. Level I model: Only the peak hour demand is being considered. Demand is considered

deterministic. When the service capacity is higher than the maximum demand rate, the level I model can be used. Also, the assumption should be made that no random fluctuations in demand. Typically, the level I models are used for planning level studies. (e.g.: design peak hour). Level II model: Time variation in the average demand is considered. But stochastic nature is not considered. Again Level II models use deterministic demand rates and service rates. Therefore, if the demand rate exceeds the maximum service rate of a facility, delays and queue will occur. Furthermore, if the average demand rate just below the maximum capacity rate at a server, these models estimate no delays. (e.g. accounting type model). Level III model: This model is based on the steady-state queuing analysis. Therefore, the average demand rate and service rate of each component remain constant at any given time (e.g. Queuing theory-based model). Level IV model: Both average service rates and demand rates are probabilistic and explicit functions of time as well[3]. (e.g. Time-varying probabilistic queuing).

Mostly, the queuing theory is used for the analysis of queues or waiting lines under stochastic processes. To do that analysis, few assumptions should be made under analytical modeling. The first one is a steady-state condition assumption during peak periods. But, if the demand is higher than the capacity, this steady-state assumption is not valid. Given the very sharp hourly variations in passenger demand for airport services, the assumption of steady state conditions tends to be violated very easily. Thus, the ability of using such models for evaluating bottlenecks or congestion scenarios is not possible. On the other hand, deterministic models work well when the average demand is higher than the average service rate. However, these models cannot model delay when average demand is less than service capacity. Nevertheless, the advantage of analytical models is that they can be used to make a quick analysis of queuing systems with minimal data requirements. Hence, decision makers make use of such models in planning level decision making to determine design parameters in terms of the number of service processors and queuing space requirements [2].

When it comes to Simulation, it has the capability to model more difficult systems with recent improvements in computer software. Also, it allows in producing better outcomes within a realistic time frame [2]. The research has investigated that simulation is a beneficial tool to forecast both time dependent and stochastic variation of demand occurring at airports. Furthermore, based on previous work, simulation can be used to model different passenger behaviors, staffing schedules, and changes in passenger demand depending on the day of week or time of the day[4]. Thus simulation modeling is capable of including far many number of variables and their complex dependencies which is beyond the scope of simple analytical models. Among various simulation methods available most of the research use the DES method to model airport passenger processing. The DES models are capable of modeling average waiting times and queue length as well as the related maximum and minimum values in relation to the number of resources that are operating in the model. The DES is often used to model a system characterized by difficult processes, combined with infrastructure at restricted capacity [1]. Also, once we have developed a valid simulation model, the model can evaluate different operating procedures, new policies or methods in a virtual environment without expensive experimentation using

the actual system. Further, alternative methods can be examined such as different check-in policies [5].

The simulation was used to study operational rules at the check-in counters. When it compares common versus dedicated check-in, the results concluded that common check-in shows a considerable reduction in average waiting time. When it comes to dynamic versus static opening and closing check-in policy, dynamic opening and closing check-in policy is capable to achieve less number of operating hours. Queuing time is depending on passenger distribution among the check-in counters. Therefore, queuing time and queue jumping can be avoided through the single queue for multiple check-in counter method. In addition to that, if the check-in period is extended prior to flight departure, it will result in a significantly dropped average queuing time [6].

The simulation model was applied to evaluate four cases in Santiago's international airport's baggage handling system. The first case was aimed to find the bottlenecks in the baggage carrousel system. The second case assessed the impact on the system due to certain changes in passenger queuing (e.g. assignment of the common passenger queue to check-in counters). In the third case, they evaluated the available options for balancing the baggage load at loading carrousel area. In this case, the impact of available alternatives on passenger queuing systems was evaluated (e.g. Change number of check-in counters allocation). In the final case, a simulation tool was used to analyze "what if scenarios" (e.g. Determine impacts when a major break down has occurred in the baggage carrousel system). Therefore, the Simulation model was used to evaluate complex systems in a detailed manner[7].

IV. SIMULATION MODEL DEVELOPMENT

A. Discrete Event Simulation Model Development

The simulation model is based on the DES theory. A DES can be described as a process where the state variable changes only at those discrete points in time when the event occurs [5]. It ignores what happens to the state variables at intermediate time points. Therefore, this is a very popular simulation method used to model systems that can be represented as a collection of discrete events such as logistics operations, healthcare systems, manufacturing processes, etc. Similarly, the airport is a very popular application of DES modeling. An airport can be easily modeled as a network of interconnected sub-processors for both passengers and aircraft. Passenger traffic flow at the airport terminal can be viewed as a discrete stochastic process where passengers as entities move through a series of discrete events or processors. When studying the efficiency of the passenger handling process, the ability to focus on the behavior of the key events is more important than focusing on time continuously. Therefore, the DES model well suited as a methodology to investigate airport departure passenger processing. A DES model was constructed using simulation software known as Simio 11 version. There are multiple versions available in Simio. The Simio student academic version was used for model development.

The simulation model was developed by way of several key stages. The first stage in the simulation model is to understand and define the empirical system completely including all the events, their sequences, and logical conditions. Also, the important input and output parameters were identified associated with each event in the process. The

simulation model must be able to closely imitate the empirical system under a given set of parameters. As the second stage, the DES model was developed based on key events in the process, parameters of interest, and their logical sequence. In the third stage, the collected data were input to the model and validate the model. In the final phase, the validated simulation model is used to analyze the different scenarios in the current airport system.

Fig. 1 shows the departure passenger terminal system at BIA. Passengers come to the airport using different types of transport modes. They get down from their vehicles at the curb area. They enter to the airport departure terminal through the entrance. After entering to the terminal building, passengers move to the departure lobby. The passenger has to wait there until check-in counters are opened. After opening the check-in counters, passengers move through first security screening and enter the check-in area. Two types of security screening points are established for business class passengers and economy class passengers. Then, they are directed to the check-in area and service is performed at the check-in counters based on their airline, ticket class, and type of the check-in (online or at the counter). Also, they hand over their check-in luggage to the ground handling staff. After the check-in, passengers are directed to the emigration counters with their hand luggage. After the emigration, they have to pass the departure corridor to reach the gate area. They have to wait in the waiting lounge until gates are opened. Finally, they start boarding to the aircraft through the gates. Also, transfer and transit passengers join with the OD passengers at the waiting lounge.

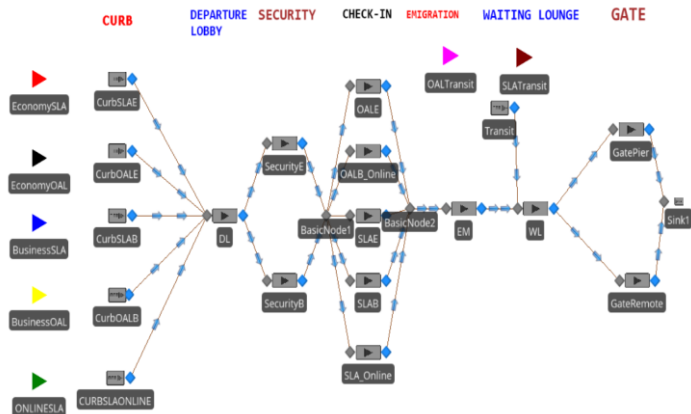


Fig. 1. Simulation model layout using Simio 11

B. Data Collection

Primary data collection methods were used including observation survey and passenger interviews at Bandaranaike International Airport. Three days of the week with the highest passenger movements were selected including Friday, Saturday, and Sunday. Data were collected within a 6-hour duration each day covering the peak period. Peak hours and peak days are determined using weekly flight schedule at BIA. Therefore, data collection was carried out in the final week of September 2019. Service time data, passenger arrival time data, and queue length data were collected at several key locations of the departure concourse. Queue Length was measured once per 10-minute time period at each server. Fig. 2 shows the locations where data was collected for the study.

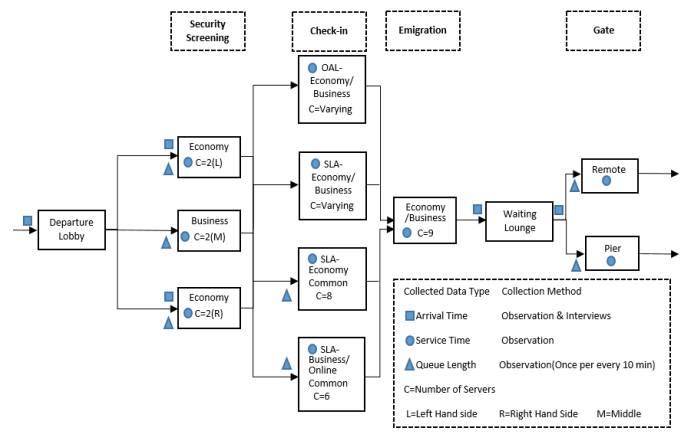


Fig. 2. Data collection in BIA departure terminal

C. Data Analysis

Using the above arrival data collection, earliness arrival distribution was found. Fig. 3 shows that most of the passengers arrive 160 to 170 minutes before their scheduled flight. Data reveals that 66% of passengers arrive at the airport less than or equal to 3 hours before the flight departure.

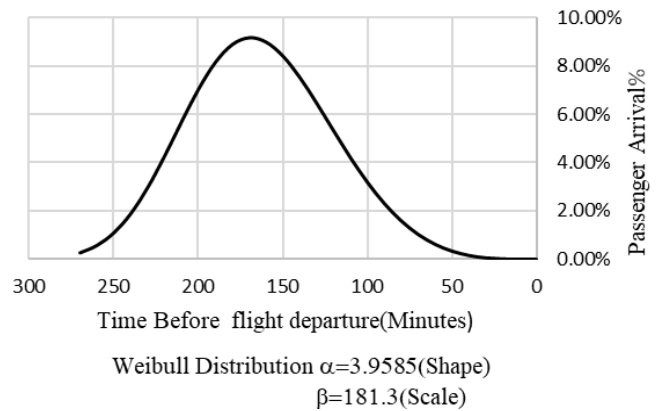


Fig. 3. Earliness arrival distribution at BIA

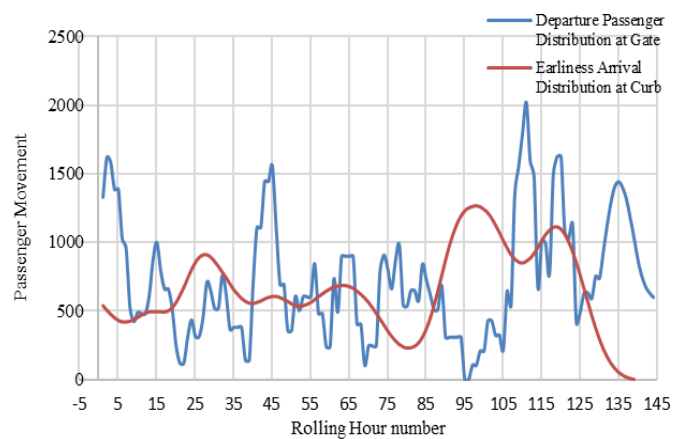


Fig. 4. Daily departure passenger variation at BIA

When it compares the queue length of the empirical system and simulation output (average, maximum and minimum queue length), the linear relationship shown in Fig. 6 was observed. The correlation coefficient between the simulation results and empirical values is 0.998. It is a positive relationship. For the hypothesis test, p value equals 0.000. This p value is less than 0.05. Therefore, it can be concluded that the relationship is statistically significant. Based on the gradient of the observed relationship in Fig. 6, the simulation model seems to overestimate the queue length parameters. This systematic overestimation could be due to the higher average load factor (75%) used for estimating the passenger arrival rate.

VI. SIMULATION RESULTS AND DISCUSSION

Fig. 7 shows that the economy class security screening and check-in processes are bottlenecks for SLA economy class passengers. The SLA economy class passengers have to wait 4 minutes to 16 minutes in the queues at the security screening. When it considers the check-in, Fig. 8 shows that departure passengers have to wait for an average of 30 minutes in the queues. Fig. 8 shows that the economy class security screening and check-in processes are bottlenecks for OAL economy class passengers. The OAL economy class passengers have to wait for an average of 16 minutes in the check-in queues.

Fig. 9 shows the processing time distribution of the SLA(Economy class) passenger throughout the departure process. SLA (Economy class) passengers have to wait 24% of time in the check-in queues from their total processing time. Also, SLA passengers spend 51% of the time at the waiting lounge and 12% at the departure lobby from their total processing time (Concession area, Transit area etc.).

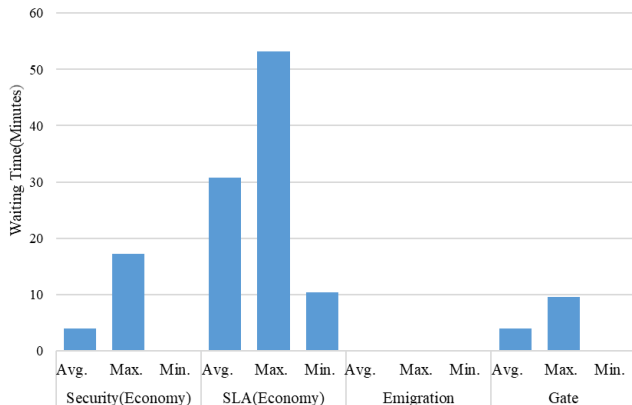


Fig. 7. Waiting time distribution of SLA passengers

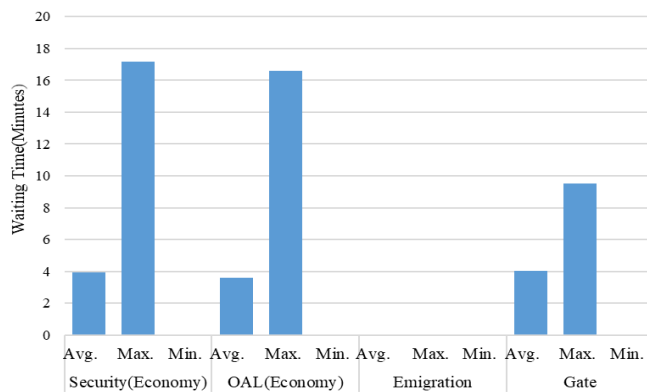


Fig. 8. Waiting time distribution of OAL passengers

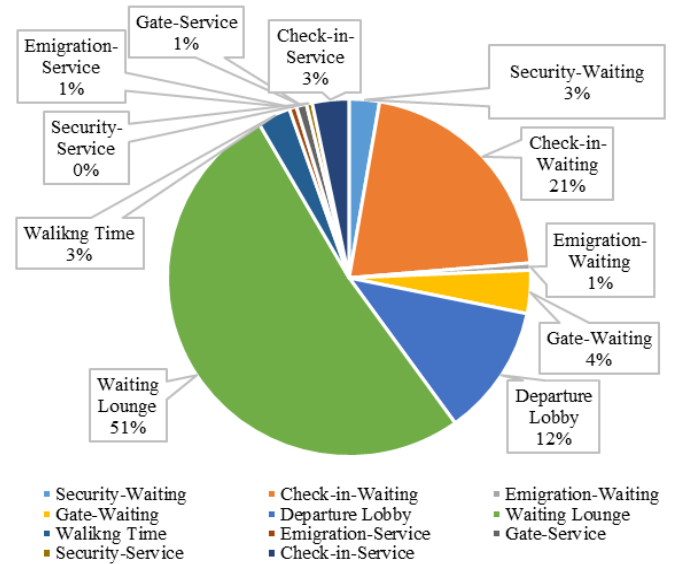


Fig. 9. SLA-Economy class passenger processing time

VII. WHAT IF ANALYSIS

A. Case I: Implement a Common Server Policy for Security Screening

According to the current layout at BIA, separate security screening points are allocated for business class passengers and economy class passengers (Two magnetometers for business class and four magnetometers for economy class passengers). The simulation model was used to investigate the impact, if all security screening points are allocated for common usage assuming passenger demand distribution and service time distribution are same for both cases. Fig. 10 shows that common server policy cause to decrease the waiting time at the security screening. Therefore, if the common server policy is going to be implemented at security screening, the capacity of the check-in should also be increased. Because the throughput of the security screening increased under common server policy. If not, waiting time at check-in could be increased due to higher passenger demand to the check-in.

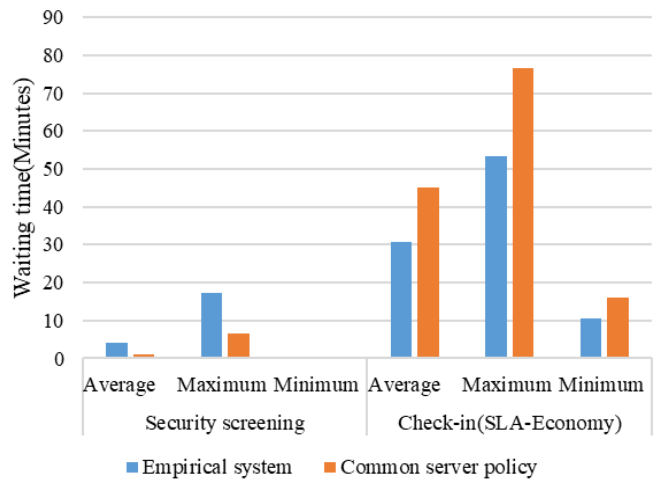


Fig. 10. Case I: Waiting time distribution at security screening and check-in

B. Case II: Increase the Number of Check-in Counters at the Srilankan Airlines' Common Check-in Area

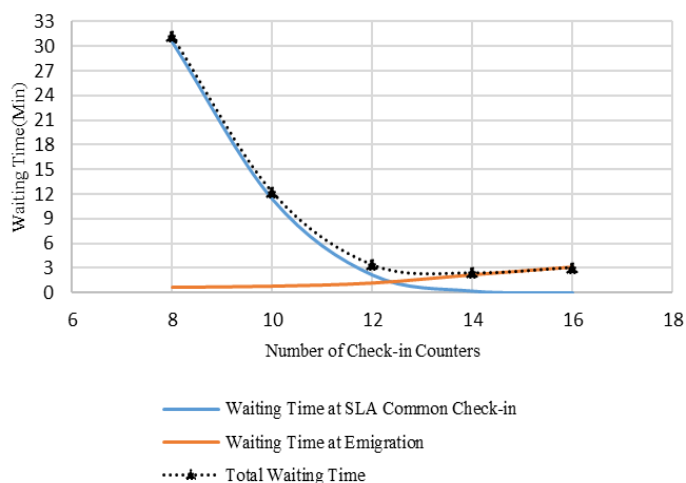


Fig. 11. Average waiting time at SLA common check-in and emigration

The validated simulation model can be used to determine the impact of changing the number of check-in desks on waiting time. The waiting time variation at the SLA common check-in can be shown as Fig. 11. There are about 8 check-in counters are allocated for SLA economy class check-in passengers. The simulation model was used to analyze waiting time variation against the number of check-in desks, assuming all the parameters including service time distribution and passenger demand variation were same for all scenarios.

This analysis emphasizes that the waiting time is decreasing exponentially when the number of check-in desks is being increased linearly. However, the average waiting time at the emigration was increased as shown in Fig. 11 due to higher passenger demand. Because of that, when the number of check-in desk allocation increases, the throughput of the check-in also increased. Further, the optimum number of check-in counters was determined using the total waiting time graph. According to the total waiting time graph, 13 common SLA check-in counters are required to maintain minimum total waiting time.

VIII. CONCLUSION

In this study, only the departure passenger terminal was considered due to the critical nature of processes such as security screening, check-in, emigration, and gates. The developed simulation model was a complete flow path to analyze the departure passenger flow. The Simio 11 software was flexible enough to model most of the attributes of the departure processing at an international airport. Also, the validated simulation model is capable to analyze the different

scenarios in the system without investing on the infrastructures. Mainly, the validated model identified the existing bottlenecks such as check-in and security screening processes. The current number of SLA common (economy class) check-in counters are not capable to handle the existing passenger demand. Therefore, it is better to allocate at least 13 counters for the SLA common economy class check-in. Also, the waiting time can be decreased by changing the existing policies in the system.

Some assumptions were made during the model development and analysis. The load factor of the aircraft was assumed as 75% and 30% of the passengers were assumed as the transit passengers. As another assumption, it was assumed that 9 emigration counters were opened out of 19 counters during the peak period. The consideration of rolling hours for analysis was more accurate than the clock hours. Because there might be a peak hour between two clock hours.

Limitations of the data collection process such as the limited number of permissions to enter the airport and limited areas to access reduce the number of data samples in the study. Also, the results of the study might be affected due to these limitations. Also, the passenger flow of the days which was selected for the data collection might be different from the predicted passenger flow of those days. Because, due to the Easter Sunday bomb attack and its prevailed circumstances in the country, some slight changes were occurred in the flow of passengers compared to the actual passenger flow. Also, the parameters calculated in the study such as waiting time and service time of the BIA could differ according to the efficiency of the staff. These facts might affect to the final result of the analysis.

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